AIFFELTHON · SPATIAL AVLA · POC BASELINE

AVLA: Audio–Vision–Language Alignment

A contract-first LangGraph DAG that turns siren detection, DOA tracking and visual anomaly into one VLM-ready context frame for a driving operator.

> **Contribution** — a schema-validated multi-agent pipeline ($S_I \rightarrow [S_A \parallel S_T \parallel S_V] \rightarrow S_B \rightarrow S_C$) with a live 2×2 operator UI, measured on real dashcam and real siren recordings — and the honest limits that come with it.

3.06
ms p50 UI fast path (target < 33 ms)

0
schema errors · All_OK = True

0.827
siren head accuracy (mixed clips)

0.70
trigger rate on real sirens

Problem
Sirens are heard *and* seen

› **AVLA PoC**
DAG · contracts · hysteresis

› **Operator**
2×2 view · ALERT_OPERATOR

1 · CONTRACT
every inter-agent edge is schema-validated — the DAG aborts loudly instead of drifting silently; All_OK is the only accepted pass state.

2 · REAL DATA
the synthetic siren baseline was **not representative** — the head is re-trained on real G44 sirens: 100% test accuracy on C01 (>99.8% confidence), best epoch 13/30, loss 0.6895 → 0.0016.

3 · STABILIZE
2 s window / 1 s hop, **5 consecutive** frames above –0.04 — the streak rule removes exactly the T09 isolated-threshold false positive.

4 · GATED OUTPUT
S_C hands the event to Qwen2.5-0.5B at most once per 5 s; a VLM failure surfaces in the status panel — it **recommends, never commands**.

5 · HONEST LIMITS
vision runs 242–301 ms/frame (off the fast path); trigger rates are **provisional**; PoC stages are mock logic — no vehicle control is produced.

Repo **spatial-avla-pipeline** · report **docs/AVLA_PoC_Final_Report.pdf**
All numbers are project measurements (outputs/*.json, abl_report_s_*.json, p_siren_report.json).
Figures rendered from project artifacts; no project file was modified.

00 · IN ACTION
The 2×2 operator view

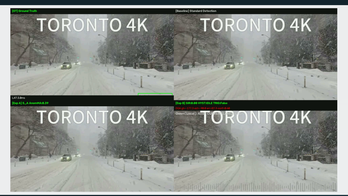


Fig. 0. Screenshot from the running Gradio app on clip C01 (day) — a real multi-view render, not a mock-up. Bottom-right tile: the operator recommendation.

01 · PROBLEM
One event, three modalities

A siren is **audible** (direction of arrival), **visible** (surrounding traffic, weather) and must become **language** an operator can act on.

Single-modality detectors fail where it matters: rain masks the siren; a wet tunnel wall looks anomalous.

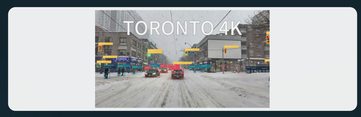


Fig. 1. S_A on a real frame (C01): 41 RT-DETR-L detections.

02 · SYSTEM
LangGraph DAG, six agents



S_A vision anomaly · S_T siren head · S_V verification fixture · S_B fusion + 35-frame hysteresis · S_C VLM payload.

03 · CONTRACT
Schema-first, not prompt-first

Every stage exchanges one **FrameAudioPacket** validated against a Draft-7 JSON Schema — **pre** and **final**. A violation is an error, not a warning.

Result: **All_OK = True**, **Errors_Count = 0** across the PoC run.

04 · LATENCY
Fast path measured



Fig. 2. UI fast path: p50 3.06 ms, p95 3.85 ms (target < 33 ms).

05 · SYNC
Audio and video agree

Frame-exact audio–video sync verified on **4 clips** (T09, C01, C02, C04), 450 frames each at 16 kHz / 30 fps — expected sample count matched exactly.

PASS 4/4 clips, 1 800 frames

06 · VISION (S_A)
RT-DETR-L + MAE



Fig. 3. Mean anomaly score by condition vs the 0.20 mock baseline.
Rain roughly **doubles** the day score (0.249 → 0.498): the detector reacts to a wet tunnel wall.

07 · COST
Why vision is off the fast path

RT-DETR-L inference costs **242–301 ms/frame** on this machine, so S_A runs **off** the 33 ms UI path; the fast path keeps the contract and hysteresis logic.

The live monitor is therefore throttled to **0.5–1.25 FPS**, matching the measured checkpoint speed.

08 · AUDIO (S_B)
WavLM head + hysteresis

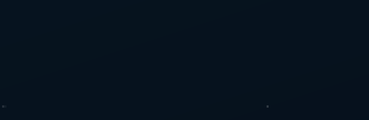


Fig. 4. Trigger rate by condition, with and without hysteresis.
Real sirens trigger on **0.70** of frames, ordinary day traffic on **0.027**.

09 · HEAD QUALITY
Where the misses are

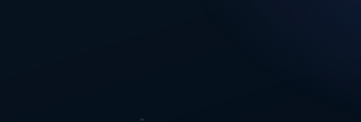


Fig. 5. recall 0.781, specificity 0.961, accuracy 0.827.
fn 1 902 vs fp 116 — misses dominate because the negative clips are long; hysteresis turns isolated positives into a held event.

10 · REAL VS SYNTHETIC
The baseline was wrong

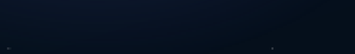


Fig. 6. WavLM frame cosine to the synthetic siren centroid.
Real sirens sit **farther** from the synthetic centroid than siren-mixed dashcam audio — the synthetic baseline was not representative.

11 · FIX
Train on real sirens

Re-training the head on the **real G44 sirens** gives **100%** test accuracy with >99.8% confidence on C01, after restoring the best epoch (13 of 30, early-stop patience 3, loss 0.6895 → 0.0016).

WavLM-base-plus · L2-logistic head

12 · OPERATING POINT
Threshold trade-off



Fig. 7. TPR / FPR per mode (12 positive, 5 negative clips).
2 s window / 1 s hop, **5 consecutive** above –0.04. T09 crossed the single-frame threshold 6/14 times but never 5 in a row — the streak rule removes exactly that false positive.

13 · VLM PAYLOAD
Evidence to language

S_C hands the fused event to **Qwen2.5-0.5B-Instruct** in a separate background task and surfaces **MONITOR / CAUTION / ALERT_OPERATOR**.

Rate-limited to **once per 5 s**; a Qwen failure appears in the status panel instead of being swallowed. It recommends — it never commands.

14 · EVALUATION
What was measured

Dashcam: C01 day, C02 rain, C04 night, T09 untouched — 450–451 frames @30 fps, 16 kHz stereo.
Real sirens: 6 G44 clips, 8 600+ frames.
Audio templates: 12 positive / 5 negative clips.
Sync: 4 clips, frame-exact.

15 · HONEST LIMITS
What this does not prove

- AST template clips overlap the evaluation set — **not independent**.
- Trigger rates are **provisional operating points**.
- Vision runs **242–301 ms/frame**, off the fast path.
- Clip-level scores read the **whole clip** audio — not causal real-time inference.
- PoC stages are **mock logic**; **no vehicle control** is produced.

16 · NEXT
From PoC to deployment

held-out real sirens › GPU live inference › 3D vision DETR3D → PETR

Build a **real, independent** siren split, move RT-DETR-L onto the live path, and port the visual stage to **3D detection** (DETR3D → PETR → Sparse4D).